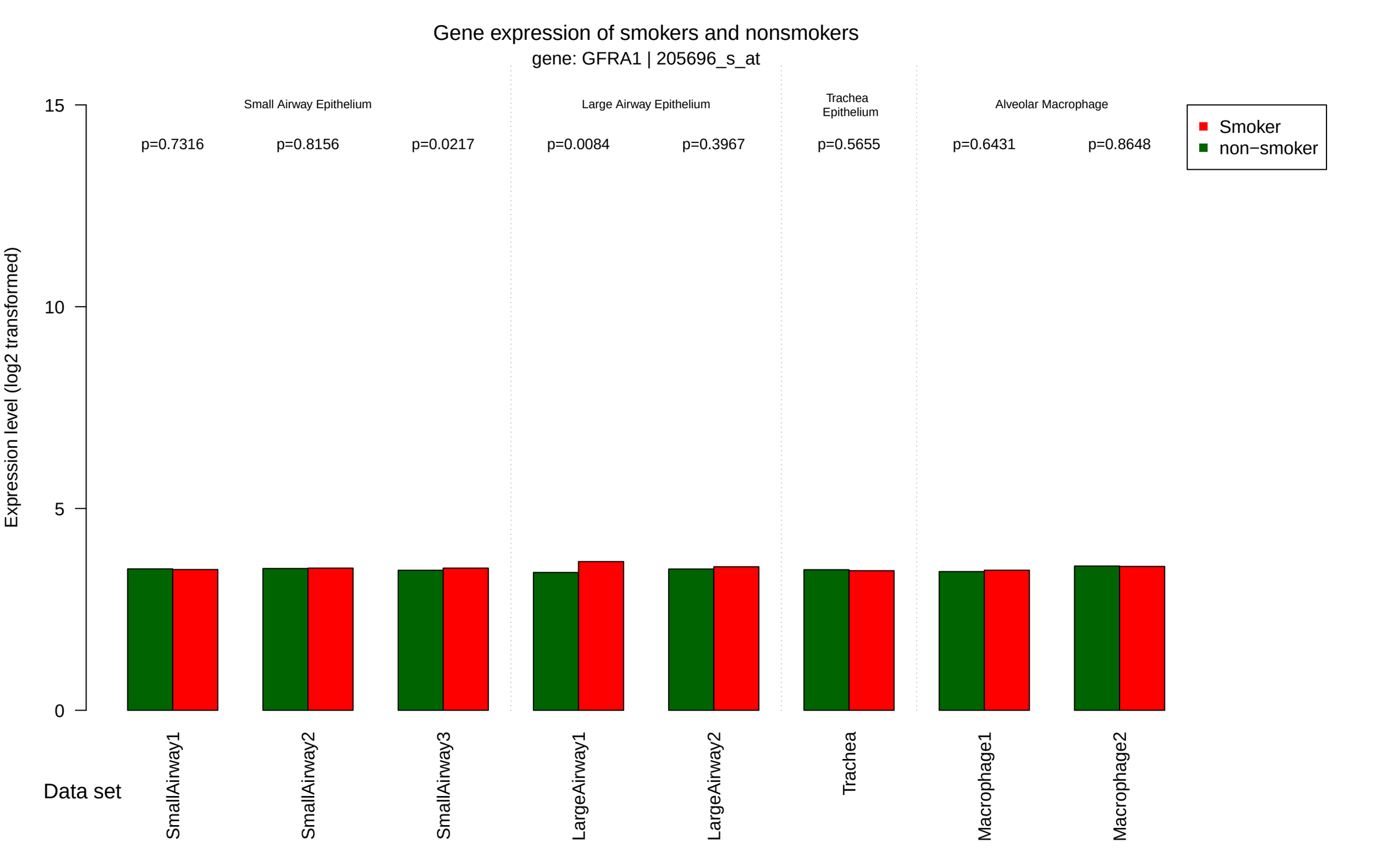
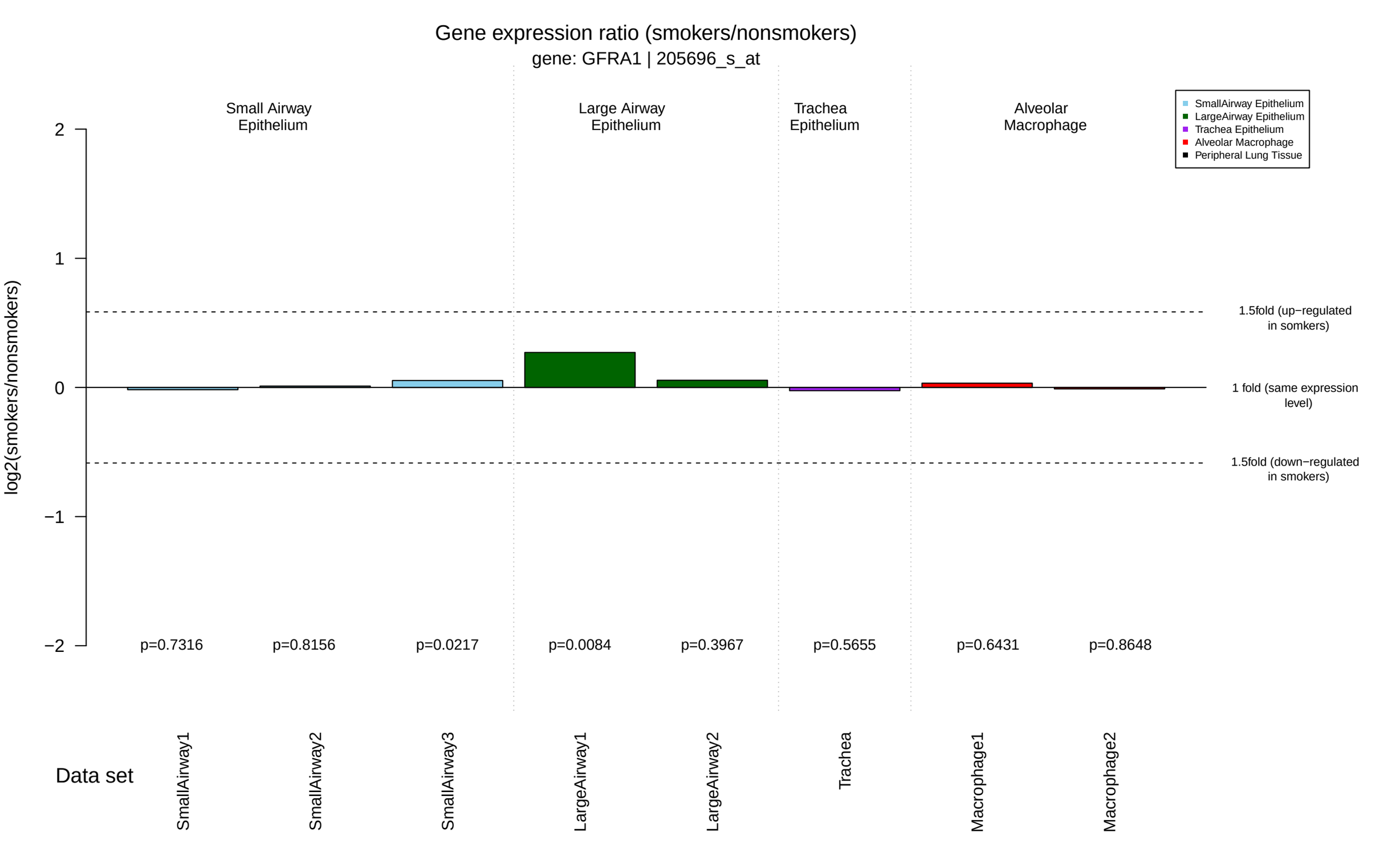
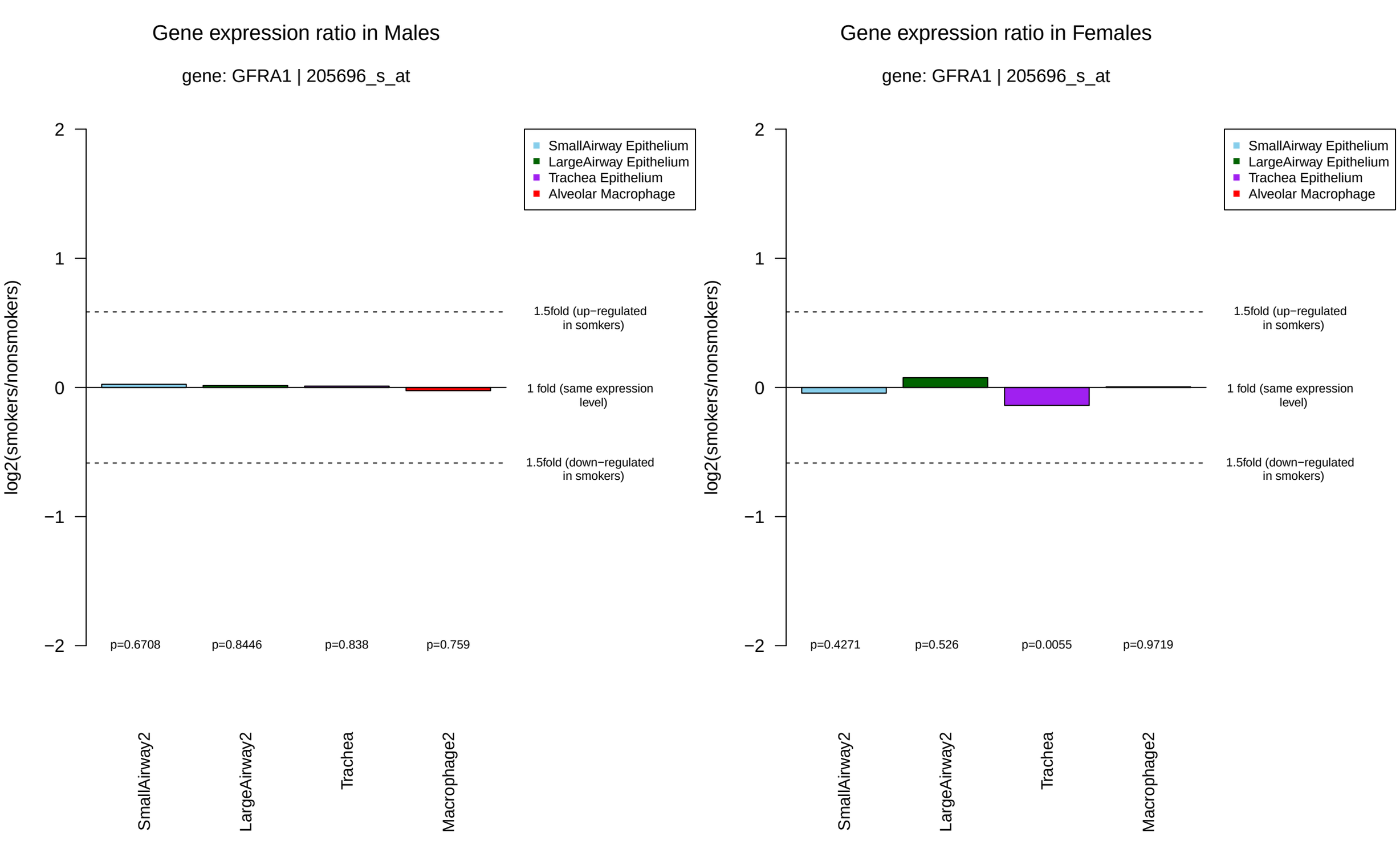
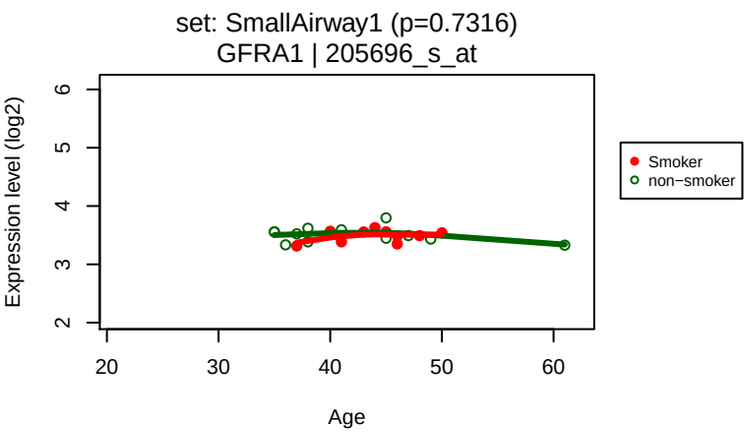
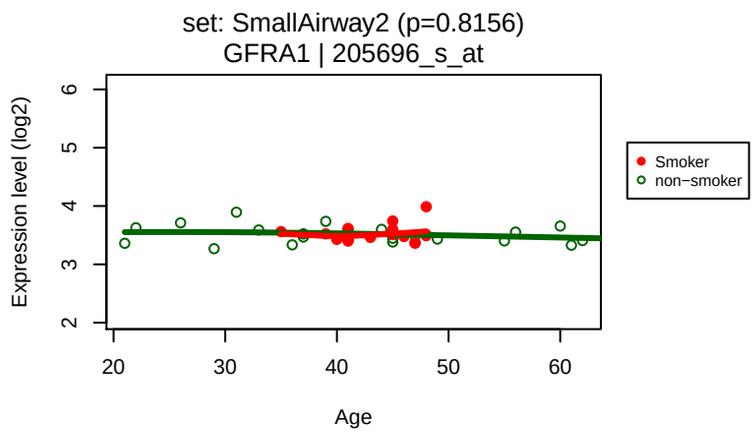
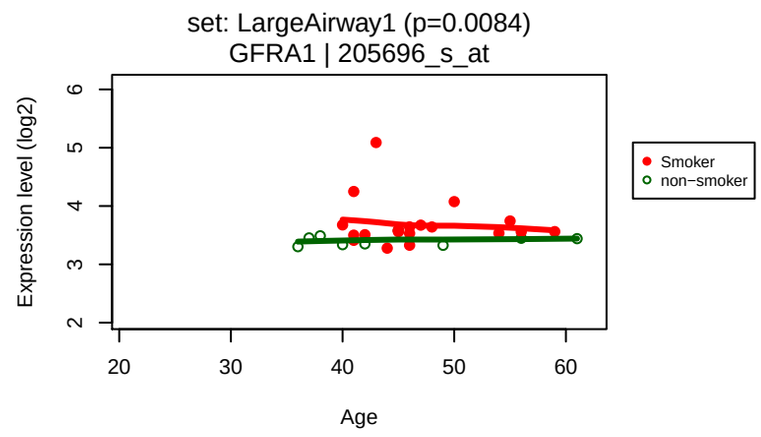

Gene expression of smokers and nonsmokers
gene: GFRA1 | 205696_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7316
p=0.8156
p=0.0217
p=0.0084
p=0.3967
p=0.5655
p=0.6431
p=0.8648
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

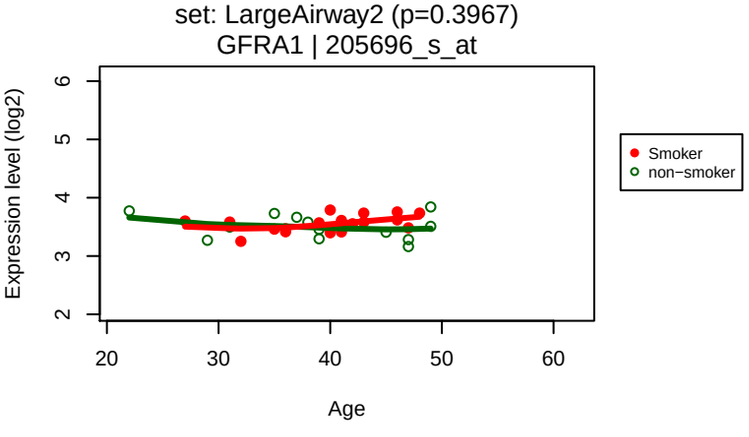

Gene expression ratio (smokers/nonsmokers)
gene: GFRA1 | 205696_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7316
p=0.8156
p=0.0217
p=0.0084
p=0.3967
p=0.5655
p=0.6431
p=0.8648
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

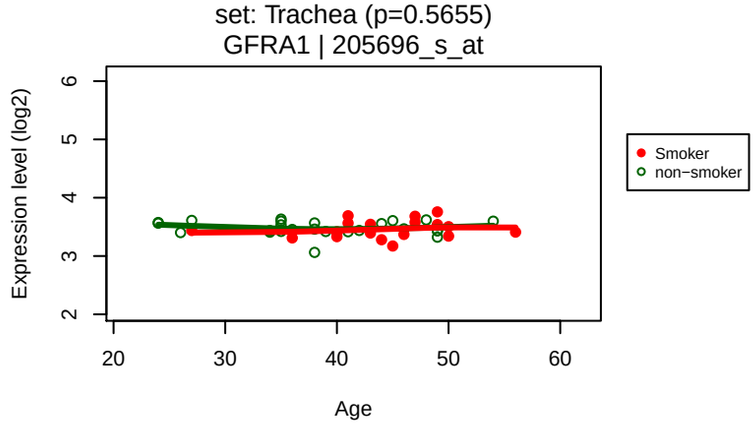

Gene expression ratio in Males
gene: GFRA1 | 205696_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6708
p=0.8446
p=0.838
p=0.759
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GFRA1 | 205696_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4271
p=0.526
p=0.0055
p=0.9719
SmallAirway2
LargeAirway2
Trachea
Macrophage2

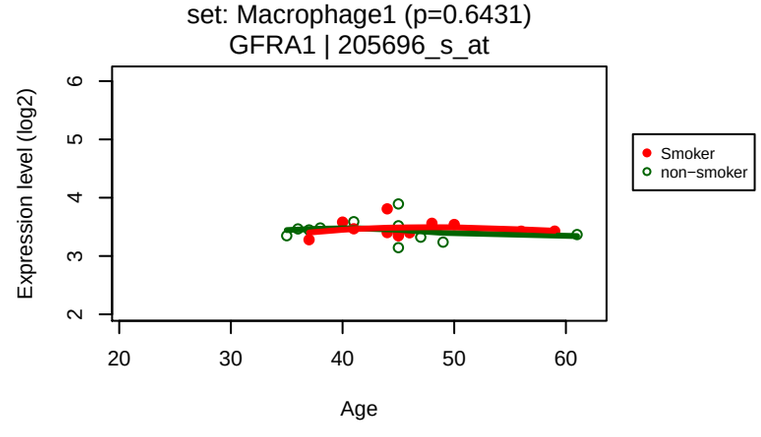

set: SmallAirway1 (p=0.7316)
GFRA1 | 205696_s_at
Expression level (log2)
Age
Smoker
non−smoker

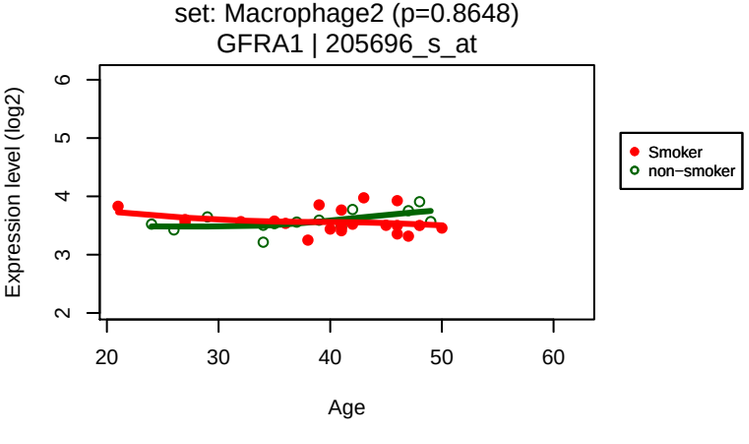

set: SmallAirway2 (p=0.8156)
GFRA1 | 205696_s_at
Expression level (log2)
Age
Smoker
non−smoker

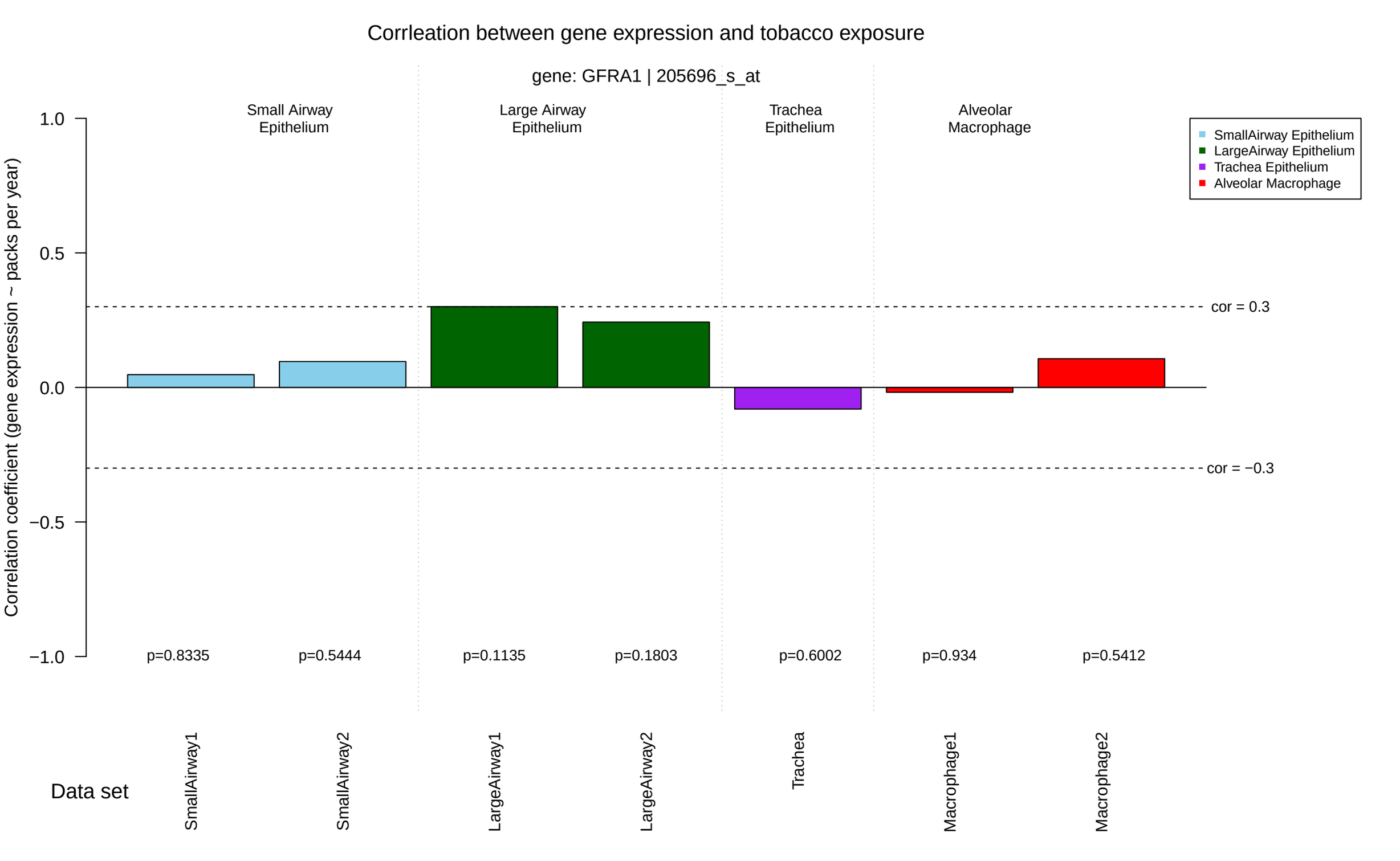

set: LargeAirway1 (p=0.0084)
GFRA1 | 205696_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3967)
GFRA1 | 205696_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5655)
GFRA1 | 205696_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6431)
GFRA1 | 205696_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8648)
GFRA1 | 205696_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GFRA1 | 205696_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8335
p=0.5444
p=0.1135
p=0.1803
p=0.6002
p=0.934
p=0.5412
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2